

MAT 167 Final Project

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Classical Multidimensional Scaling of Major U.S. Airports: Geographic
Reconstruction and Operational Similarity

Hand Written Report

Classical Multidimensional Scaling of Major U.S. Airports: Geographic Reconstruction and Operational Similarity.

MAT 167.

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1. Motivation

I am majoring in Data Science and minoring in Econ. In the future, I am interesting in working in airport management or for an airport authority. Airports produce large amounts of data. These data include passenger volume, flight activity, delays, and Cancellations. It can be difficult to compare airports when several variables are considered at the same time period.

Classical multidimensional scaling (MDS), provides a way to represent distance or dissimilarity data in a lower-dimensional space. In this project, I use classical MDS to study 10 major U.S. airports. First, I test whether airport locations can be reconstructed from pairwise geographic distance alone. Second, I use operational data to compare the similarities and differences among the airports.

This topic connects linear algebra with data science and airport operations. I also shows how eigenvalues, eigenvectors, and Gram matrices can be used to solve a practical visualization problem.

2. Background from the course

The first idea is eigenvalues and eigenvectors. For a square matrix A , a nonzero vector v is an eigenvector if $Av = \lambda v$. The λ is the eigenvalue. An eigenvector stays in the same span after the matrix transformation. Its length may change, and its direction may be resered.

The second idea is inner products. For two real vectors v and w , the standard inner product is $\langle v, w \rangle = v^T w$. The inner product can be used to find vector lengths and angles. Two vectors are orthogonal when $\langle v, w \rangle = 0$.

Another related course topic is PCA. In PCA, the data are first centered. Eigenvectors are then used as new coordinate directions. The first direction keeps the most Variation in the data. The next direction keeps as much of the remaining variation as possible and is orthogonal to the first direction.

classical MDS uses a similar idea. It uses eigenvalues and eigenvectors to create new coordinate directions. However, instead of starting with the original data coordinates, classical MDS starts with pairwise distance.

fill up: This project also we eigendecomposition of a real symmetric matrix $A = V \Lambda V^T$. The column of V are orthonormal eigenvectors. The diagonal entries of Λ are the eigenvalues. In this project, eigendecomposition is used to find the main coordinate directions from a matrix created from distance data.

• New Mathematics (MDS)

1. Pairwise Distance and the Distance Matrix

Classical MDS starts with pairwise distances. Given two points x_i, x_j . Their Euclidean distance is

$$d_{ij} = \|x_i - x_j\|. \text{ If the points have two coordinates, then } d_{ij} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2}.$$

All pairwise distances can be stored in a distance matrix D . The entry d_{ij} gives the distance between x_i and x_j . The matrix is symmetric because $d_{ij} = d_{ji}$. The diagonal entries are zero because the distance from a point to itself is zero.

2. Centering the Points

Before recovering the coordinates, classical MDS centers all points around the origin. Suppose there are n points, the mean position is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. Each centered point is found by subtracting the mean $x_i^c = x_i - \bar{x}$. Then, the average of all centered points is zero. $\frac{1}{n} \sum_{i=1}^n x_i^c = 0$. Centering does not change the distance between two points. So, $\|x_i^c - x_j^c\| = \|x_i - x_j\|$. This means that the whole group of points can be moved without changing its shape or pairwise distances.

Centering matrix is $J = I - \frac{1}{n} \mathbf{1}\mathbf{1}^T$, I is the identity matrix, $\mathbf{1}$ is a vector of ones.

If x contains the original point coordinates, then the centered coordinate matrix is $X_c = JX$. Multiplying by J subtracts the mean coordinate from every point. Classical MDS uses centered coordinates because distances alone do not show the absolute location of the points.

After the points are centered, their inner products can be stored in a Gram matrix.

3. Gram Matrix

Given the rows of X are the coordinates of the points. The Gram matrix is $B = XX^T$. Each entry of B is $b_{ij} = x_i^T x_j$. This entry is the inner product between point i and point j . The diagonal entry is $b_{ii} = x_i^T x_i = \|x_i\|^2$. So, it gives the squared distances from the points to the origin.

The Gram matrix is symmetric because $b_{ij} = x_i^T x_j = x_j^T x_i = b_{ji}$.

The eigenvalues of a true Gram matrix are nonnegative.

In classical MDS, the coordinates are unknown. We should recover B from the distance matrix.

4. Double Centering

The main idea of classical MDS is to get coordinates from distance. We first connect distance with the inner product. For two points x_i and x_j , $d_{ij}^2 = \|x_i - x_j\|^2$. we expand it, we get $d_{ij}^2 = \overset{(d_{ii})}{x_i^T x_i} + \overset{(d_{jj})}{x_j^T x_j} - 2 \overset{(2d_{ij})}{x_i^T x_j}$. The term $x_i^T x_j$ is the same type of value that appears in the Gram matrix. So if we know all the distances, we can use them to recover the Gram matrix. First, we square every value in the distance matrix. Then we use the centering matrix $J = I - \frac{1}{n} \mathbf{1}\mathbf{1}^T$. The Gram matrix can be found by $B = -\frac{1}{2} J D^2 J$. D^2 means every distance in D is squared. This step is called double centering because the squared distance matrix is centered from both sides. The main purpose is to remove the extra terms that only depend on each individual point. What is left gives the inner-product information between the points. It is useful because it means we do not need the original coordinates. If we only know the distances between the points, we can still recover a Gram matrix and use it to build new coordinates.

5. Recovering Coordinates from Eigenvalues and Eigenvectors

After finding the Gram matrix B , the next step is to recover coordinates from it. Since B is symmetric we can use eigendecomposition: $B = V \Lambda V^T$, the column of V are the eigenvectors of B , the diagonal entries of Λ are the eigenvalues. In classical MDS, the largest positive eigenvalues are the most important. They show the main directions of the reconstructed points. If we want a two-dimensional result, we keep the two largest positive eigenvalues and their eigenvectors.

let V_2 contain the first two eigenvectors, and let Λ_2 contain the first two eigenvalues. The new coordinates are $X_{\text{MDS}} = V_2 \Lambda_2^{1/2}$. The square root of the eigenvalues gives the scale of each direction. The eigenvectors give the directions of the new coordinate system. If the first two eigenvalues contain most of the information, then the two-dimensional MDS map is a good summary of the original distance.

6. Rotation and Reflection

The reconstructed MDS coordinates may not look exactly the same as the original coordinates. The whole map can be rotated or reflected.

This is not a problem because rotation and reflection do not change the distances between points. The shape stays the same even if the direction of the map changes.

⇒ (next page)

For example, two points that are close before the transformation will still be close after it. The same is true for points that are far apart.

Because classical MDS only uses distance information, it cannot know the original left-right or up-down direction of the points. It only tries to keep the pairwise distances.

So when we compare an original map with an MDS map, the most important thing is whether the distances and relative positions are preserved.

4. Computational Exploration

1. Experiment 1: Geographic Reconstruction

In the first experiment, I wanted to see if classical MDS could rebuild the locations of airports using only pairwise distances. I used 10 U.S. airports: BOS, JFK, ATL, MIA, ORD, DFW, DEN, LAX, SFO and SEA. I started with their latitude and longitude values. I then converted them into approximate planar coordinates in km.

Next, I calculated the Euclidean distance between every pair of airports. These distances were stored in a 10×10 distance matrix D .

Then I squared the distance matrix and used the centering matrix. $J = I - \frac{1}{n} \mathbf{1}\mathbf{1}^T$.

I used double centering to calculate $B = -\frac{1}{2} J D^2 J$.

The matrix B was symmetric, and the sum of each row was approximately zero. I also compared this matrix with the Gram matrix calculated directly from the original coordinates. The maximum difference was only 2.79×10^{-9} . This showed that double centering recovered the Gram matrix very accurately.

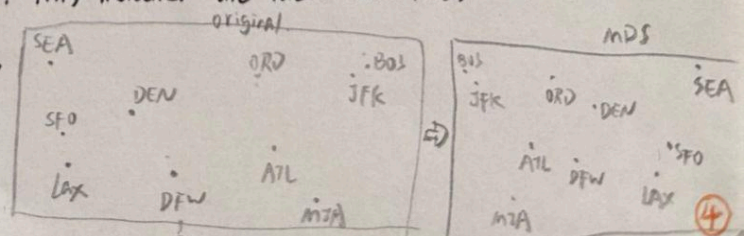
After that, I found the eigenvalues and eigenvectors of B . Only the first two eigenvalues were positive and important. They were about 27,556,003 and 4,083,007. I used these two eigenvalues and their eigenvectors to create the two-dimensional MDS coordinates.

Finally, I compared the original distances with the distances from the reconstructed MDS coordinates. The largest difference was 5.00×10^{-12} . It is extremely close to 0. It means that the two-dimensional MDS result preserved the original distances almost exactly.

The reconstructed map was reflected compare with the original. However, the relative positions and distances between the airports were the same. This matches the idea that MDS can recover the geometry up to rotation or reflection.

latitude/longitude $\rightarrow$ planar coordinates $\rightarrow D \rightarrow D^2$

$\rightarrow B \rightarrow$ eigendecomposition $\rightarrow X_{\text{MDS}}$



2. Experiment 2: Operational Similarity of Airports

In the second experiment, I used MDS to compare the operational similarities of the same 10 airports. This time, I did not use geographic locations. I used four variables: passenger volume, departures, delay rate, and cancellation rate.

The four variables had different scales. So, I standardized each variable first: $z = \frac{x - \bar{x}}{s}$.

After standardization, each variable had mean 0 and standard deviation 1. This made the four variables more comparable.

Then I calculate the Euclidean distance between every pair of airports using the standardized values. These distances formed the operational distance matrix D_{ops} . A smaller distance means two airports have more similar operational patterns. A large distance means they are more different.

The closest pair was ORD and DEN. $d(ORD, DEN) \approx 0.765$. The most different pair was ATL and MIA. $d(ATL, MIA) \approx 4.845$. so ORD and DEN were the most similar pair, while ATL and MIA were the most different in those four variables.

After getting the distance matrix, I used double centering: $B_{ops} = -\frac{1}{2} J D_{ops}^2 J$

Then I found the eigenvalues and eigenvectors of B_{ops} . I kept the first two dimensions and used them to create the two-dimensional MDS coordinates.

The first dimension kept about 54.0% of the information, and the second kept about 36.3%.

$$54.0\% + 36.3\% \approx 90.3\%$$

This means the two dimensions kept most of the information from the four operational variables.

(raw data $\rightarrow$ standardize $\rightarrow D_{ops} \rightarrow B_{ops} \rightarrow$ eigendecomposition $\rightarrow$ 2D MDS map)

5. Discussion

The first experiment showed that classical MDS can recover the geometry of points from distances.

The reconstructed airport map did not have the same direction as the original map, but the distances were almost exactly the same. The maximum difference was only about 5.0×10^{-12} . It helped me understand that MDS focuses on relative distances, not the exact direction of the map.

The second experiment showed that MDS can be used when data are not geographic. I used passenger volume, departures, delay rate, and cancellation rate to compare the airports. ORD and DEN were the most similar pair, while ATL and MIA were the most different. The first two MDS dimensions kept about 90.3% of the information, so the two-dimensional map still represented most of the differences in the original data.

From the correlations, I think MDS_1 is mostly related to airport size and flight activity. MDS_2 is more related to delays and cancellations. On the map, ORD and DEN were close together, which matched the distance result. DFW was high on the second dimension, while LAX was lower. This shows that airports with similar size can still have different operational patterns.

I think this method is useful for data science because it turns many variables into a simple visual map. It can also be useful in airport management. An airport manager could compare airports and see which airports have similar operating patterns. This could help with benchmarking and finding airports that may have similar problems or strengths.

The limitation is that the result depends on the variables that are included. I only used four operational variables and 10 airports. Using more variables and more airports would give a different result. Also, reducing the data to two dimensions loses some information. In this project, about 9.7% of the operational information was not shown in the two-dimensional map.

6. References

Federal Aviation Administration. (2025). CY 2024 enplanements at all commercial service airports (by rank). U.S. Department of Transportation.

Bureau of Transportation Statistics. (2024). Airline on-time performance and delay data. U.S. Department of Transportation.

Bureau of Transportation Statistics. (2024). Airports statistics: domestic flights, delays, cancellations, and number of flights. U.S. Department of Transportation.

Coding and Plots Appendix

Coding & Plots Appendix — Output Results Only

All remaining code images have been removed; only output results and plots are shown below.

Airport data table

Planar coordinates (x_km, y_km)

code	airport	latitude	longitude
<chr>	<chr>	<dbl>	<dbl>
BOS	Boston Logan International Airport	42.36294	-71.00639
JFK	John F. Kennedy International Airport	40.63993	-73.77869
ATL	Hartsfield-Jackson Atlanta International Airport	33.63670	-84.42786
MIA	Miami International Airport	25.79536	-80.29012
ORD	Chicago O'Hare International Airport	41.97694	-87.90815
DFW	Dallas/Fort Worth International Airport	32.89723	-97.03769
DEN	Denver International Airport	39.86167	-104.67317
LAX	Los Angeles International Airport	33.94250	-118.40805
SFO	San Francisco International Airport	37.61881	-122.37542
SEA	Seattle-Tacoma International Airport	47.44989	-122.31178

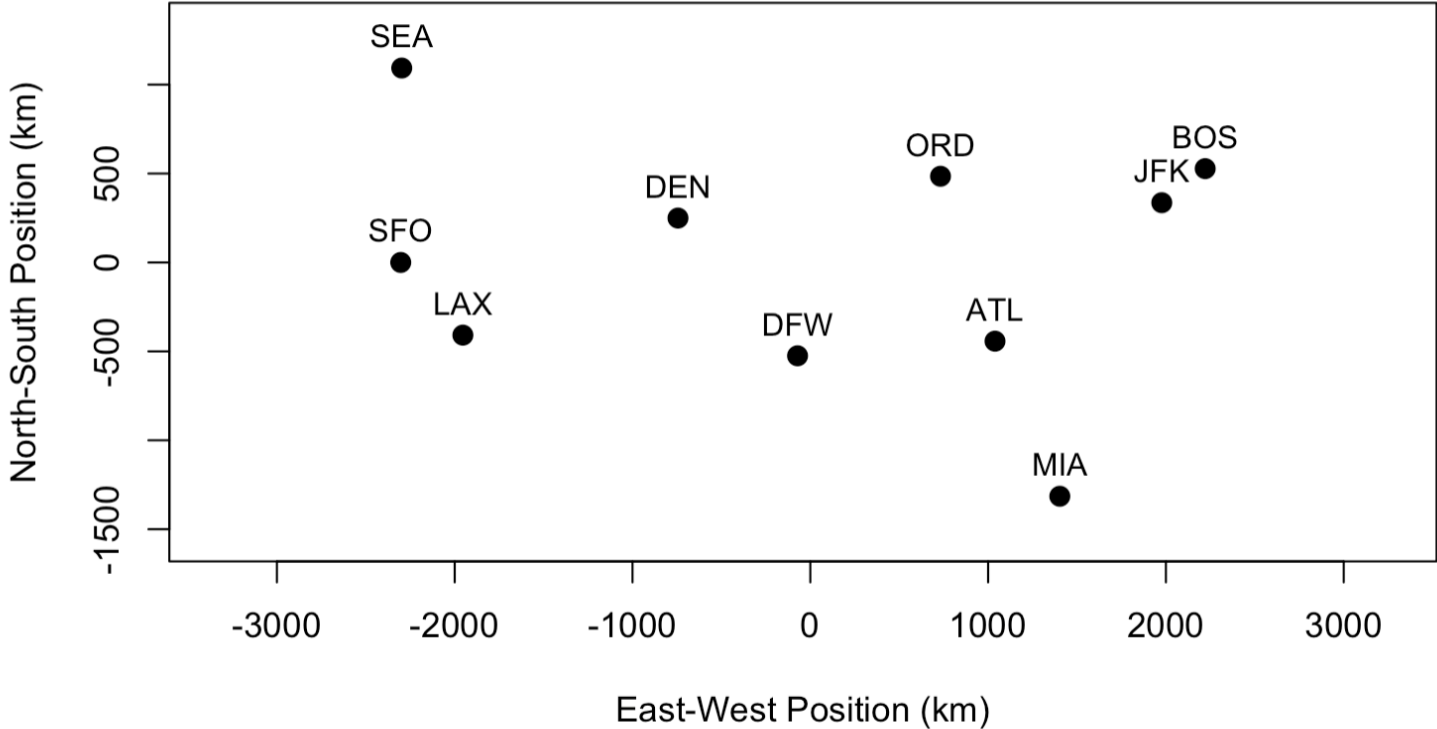
1-10 of 10 rows

code	x_km	y_km
<chr>	<dbl>	<dbl>
BOS	2220.89269	5.275919e+02
JFK	1976.71645	3.360012e+02
ATL	1038.76893	-4.427222e+02
MIA	1403.20952	-1.314639e+03
ORD	732.23565	4.846702e+02
DFW	-71.86763	-5.249472e+02
DEN	-744.37738	2.494625e+02
LAX	-1954.10510	-4.087192e+02
SFO	-2303.53912	6.778443e-02
SEA	-2297.93399	1.093234e+03

Experiment 1 — Geographic Reconstruction

Original geographic configuration of 10 U.S. airports

Original Geographic Configuration of 10 U.S. Airports



Distance matrix D (first rows shown)

	BOS	JFK	ATL	MIA	ORD	DFW	DEN	LAX	SFO	SEA
BOS	0.0	310.4	1529.4	2015.5	1489.3	2522.8	2978.3	4278.7	4555.1	4554.1
JFK	310.4	0.0	1219.1	1747.4	1253.3	2222.1	2722.5	4000.7	4293.4	4341.2
ATL	1529.4	1219.1	0.0	945.0	976.7	1113.7	1912.8	2993.1	3371.5	3673.2
MIA	2015.5	1747.4	945.0	0.0	1920.3	1673.2	2656.8	3477.4	3933.0	4415.5
ORD	1489.3	1253.3	976.7	1920.3	0.0	1290.7	1495.2	2831.0	3074.2	3090.7
DFW	2522.8	2222.1	1113.7	1673.2	1290.7	0.0	1025.7	1885.8	2292.6	2752.1
DEN	2978.3	2722.5	1912.8	2656.8	1495.2	1025.7	0.0	1377.2	1579.0	1767.9
LAX	4278.7	4000.7	2993.1	3477.4	2831.0	1885.8	1377.2	0.0	537.8	1540.8
SFO	4555.1	4293.4	3371.5	3933.0	3074.2	2292.6	1579.0	537.8	0.0	1093.2
SEA	4554.1	4341.2	3673.2	4415.5	3090.7	2752.1	1767.9	1540.8	1093.2	0.0

Distance matrix D (continued) and matrix checks

	BOS	JFK	ATL	MIA	ORD	DFW	DEN	LAX	SFO	SEA
BOS	0.0	310.4	1529.4	2015.5	1489.3	2522.8	2978.3	4278.7	4555.1	4554.1
JFK	310.4	0.0	1219.1	1747.4	1253.3	2222.1	2722.5	4000.7	4293.4	4341.2
ATL	1529.4	1219.1	0.0	945.0	976.7	1113.7	1912.8	2993.1	3371.5	3673.2
MIA	2015.5	1747.4	945.0	0.0	1920.3	1673.2	2656.8	3477.4	3933.0	4415.5
ORD	1489.3	1253.3	976.7	1920.3	0.0	1290.7	1495.2	2831.0	3074.2	3090.7
DFW	2522.8	2222.1	1113.7	1673.2	1290.7	0.0	1025.7	1885.8	2292.6	2752.1
DEN	2978.3	2722.5	1912.8	2656.8	1495.2	1025.7	0.0	1377.2	1579.0	1767.9
LAX	4278.7	4000.7	2993.1	3477.4	2831.0	1885.8	1377.2	0.0	537.8	1540.8
SFO	4555.1	4293.4	3371.5	3933.0	3074.2	2292.6	1579.0	537.8	0.0	1093.2
SEA	4554.1	4341.2	3673.2	4415.5	3090.7	2752.1	1767.9	1540.8	1093.2	0.0

[1] 310.3691

[1] 4555.081

[1] 537.7834

[1] TRUE

BOS	JFK	ATL	MIA	ORD	DFW	DEN	LAX	SFO	SEA
0	0	0	0	0	0	0	0	0	0

Squared distance matrix D^2

	BOS	JFK	ATL	MIA	ORD	DFW	DEN	LAX	SFO	SEA
BOS	0	96329	2338926	4062421	2217942	6364588	8870183	18307285	20748765	20739746
JFK	96329	0	1486156	3053524	1570835	4937929	7411841	16005967	18433439	18846038
ATL	2338926	1486156	0	893056	954019	1240275	3658731	8958451	11367086	13492749
MIA	4062421	3053524	893056	0	3687721	2799466	7058544	12092253	15468440	19496319
ORD	2217942	1570835	954019	3687721	0	1665909	2235709	8014571	9450768	9552278
DFW	6364588	4937929	1240275	2799466	1665909	0	1051980	3556327	5255998	7573883
DEN	8870183	7411841	3658731	7058544	2235709	1051980	0	1896644	2493183	3125489
LAX	18307285	16005967	8958451	12092253	8014571	3556327	1896644	0	289211	2374083
SFO	20748765	18433439	11367086	15468440	9450768	5255998	2493183	289211	0	1195045
SEA	20739746	18846038	13492749	19496319	9552278	7573883	3125489	2374083	1195045	0

Centering matrix J

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]
[1,]	0.9	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
[2,]	-0.1	0.9	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
[3,]	-0.1	-0.1	0.9	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
[4,]	-0.1	-0.1	-0.1	0.9	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
[5,]	-0.1	-0.1	-0.1	-0.1	0.9	-0.1	-0.1	-0.1	-0.1	-0.1
[6,]	-0.1	-0.1	-0.1	-0.1	-0.1	0.9	-0.1	-0.1	-0.1	-0.1
[7,]	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	0.9	-0.1	-0.1	-0.1
[8,]	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	0.9	-0.1	-0.1
[9,]	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	0.9	-0.1
[10,]	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	0.9

Recovered Gram matrix B

	BOS	JFK	ATL	MIA	ORD	DFW	DEN	LAX	SFO	SEA
BOS	5210717.5	4567346.6	2073417.7	2422784.7	1881924.8	-436568.2	-1521567.9	-4555494.7	-5115877.4	-4526683.3
JFK	4567346.6	4020304.8	1904596.4	2332026.9	1610272.0	-318444.8	-1387603.3	-4000041.9	-4553420.9	-4175035.8
ATL	2073417.7	1904596.4	1275043.8	2039630.5	546049.4	157751.9	-883678.7	-1848914.6	-2392874.9	-2871021.5
MIA	2422784.7	2332026.9	2039630.5	3697273.5	390313.6	589270.9	-1372470.7	-2204700.5	-3232437.1	-4661691.7
ORD	1881924.8	1610272.0	546049.4	390313.6	771074.2	-307050.3	-424152.6	-1628959.4	-1686700.6	-1152771.1
DFW	-436568.2	-318444.8	157751.9	589270.9	-307050.3	280734.5	-77458.0	354992.9	165514.3	-408743.2
DEN	-1521567.9	-1387603.3	-883678.7	-1372470.7	-424152.6	-77458.0	616329.3	1352631.5	1714719.3	1983251.1
LAX	-4555494.7	-4000041.9	-1848914.6	-2204700.5	-1628959.4	354992.9	1352631.5	3985578.2	4501329.9	4043578.6
SFO	-5115877.4	-4553420.9	-2392874.9	-3232437.1	-1686700.6	165514.3	1714719.3	4501329.9	5306292.5	5293455.0
SEA	-4526683.3	-4175035.8	-2871021.5	-4661691.7	-1152771.1	-408743.2	1983251.1	4043578.6	5293455.0	6475661.9

Checks: symmetry, row sums, and max |B - B_original|

[1] TRUE
BOS JFK ATL MIA ORD DFW DEN LAX SFO SEA
0 0 0 0 0 0 0 0 0 0

[1] 2.793968e-09

Experiment 1 — Eigenvalues and Reconstructed Coordinates

Eigenvalues of B

[1]	27556003	4083007	0	0	0	0	0	0	0	0
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Reconstructed 2D coordinates X_mds

	MDS1	MDS2
BOS	-2168.40	713.28
JFK	-1941.28	501.75
ATL	-1072.45	-353.41
MIA	-1509.23	-1191.43
ORD	-688.68	544.78
DFW	27.27	-529.14
DEN	762.79	185.70
LAX	1912.60	-572.30
SFO	2295.31	-194.49
SEA	2382.06	895.25

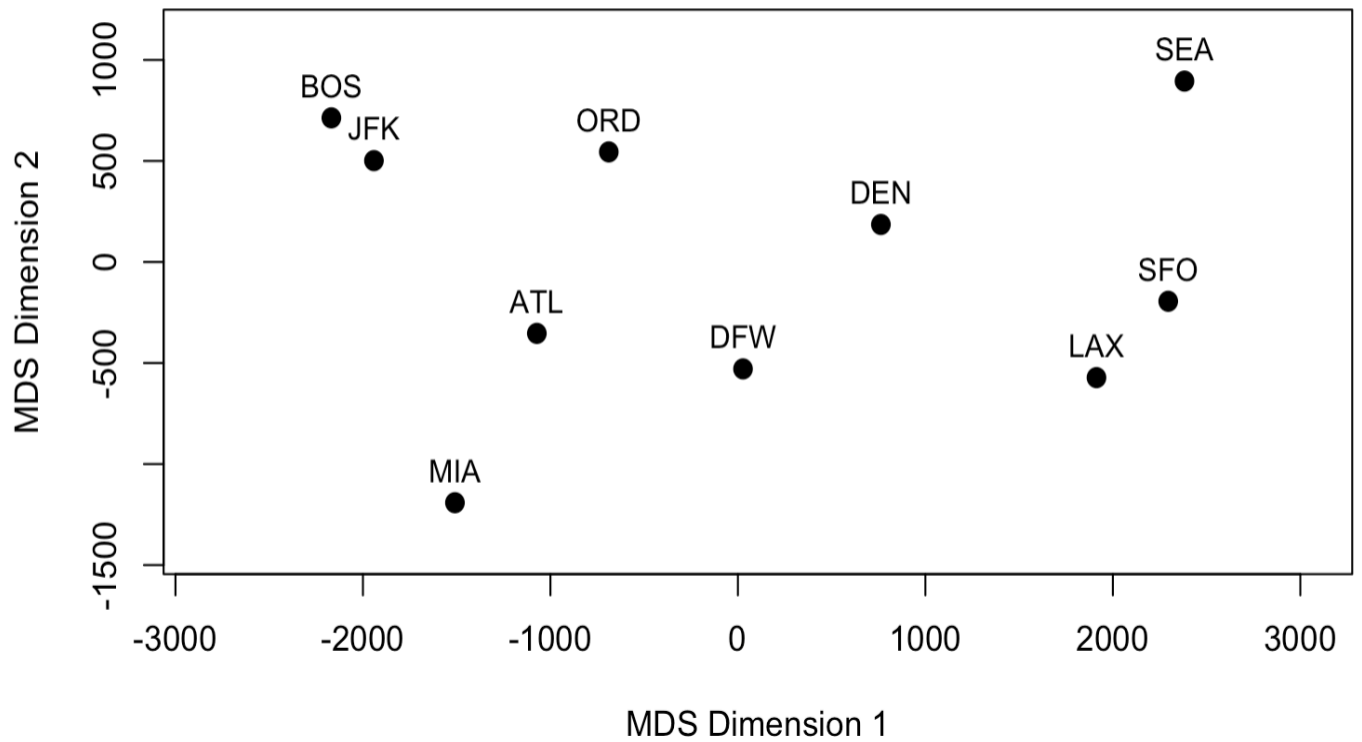
Distance preservation check

[1]	5.002221e-12
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Experiment 1 — Plots and Dimension Proportions

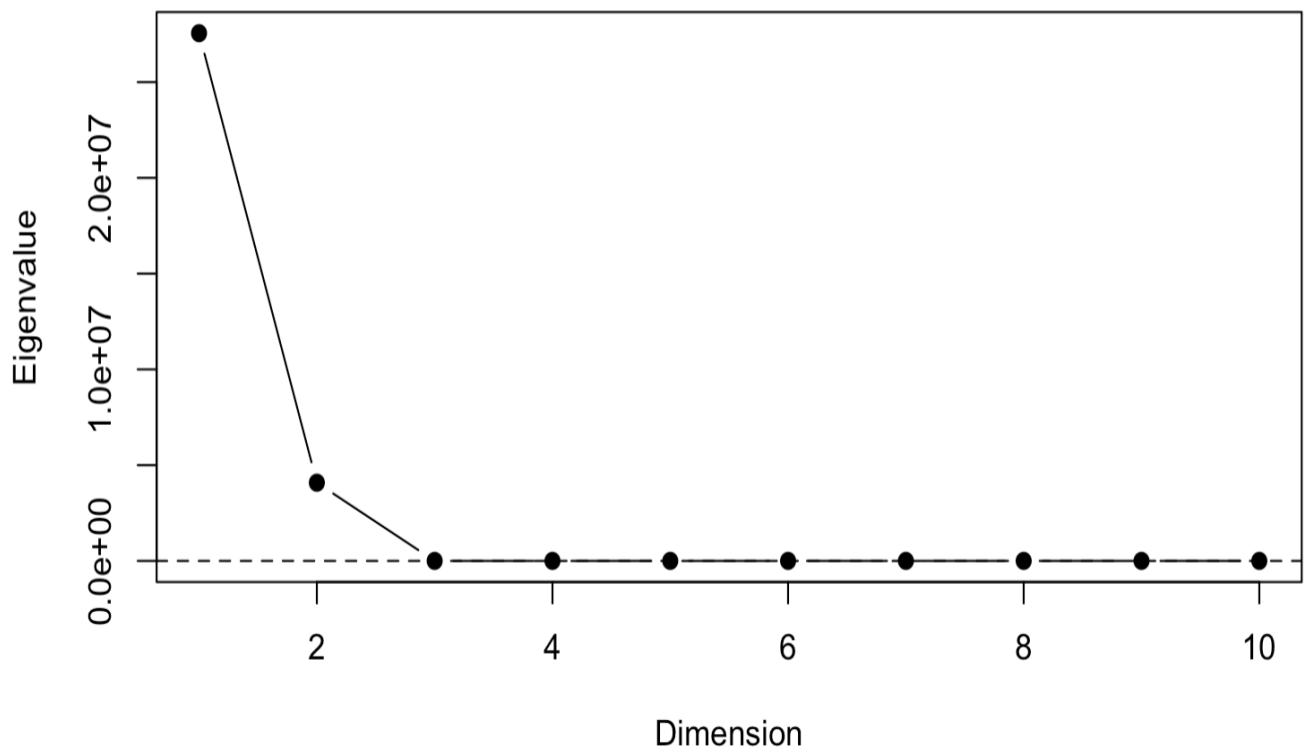
Airport locations reconstructed by classical MDS

Airport Locations Reconstructed by Classical MDS



Eigenvalues of the Gram matrix Dimension proportions

Eigenvalues of the Gram Matrix



Experiment 2 — Operational Data

Operational data table (part 1)

code <chr>	passengers <int>	departures <int>	delay_rate <int>	cancellation_rate <dbl>
BOS	21090721	143500	22	1.45
JFK	31466102	123300	22	1.86
ATL	52511402	341900	21	1.00
MIA	26588002	109900	29	1.71
ORD	38575693	280100	23	1.35
DFW	42351316	313600	29	1.86
DEN	40012895	308600	24	1.13
LAX	37760834	194100	19	0.81
SFO	25078968	135000	25	1.33
SEA	25414592	163700	22	1.11

1-10 of 10 rows

Standardized operational variables

	passengers	departures	delay_rate	cancellation_rate
BOS	-1.33	-0.76	-0.49	0.25
JFK	-0.27	-0.98	-0.49	1.38
ATL	1.88	1.45	-0.79	-1.00
MIA	-0.76	-1.13	1.65	0.96
ORD	0.46	0.76	-0.18	-0.03
DFW	0.84	1.14	1.65	1.38
DEN	0.60	1.08	0.12	-0.64
LAX	0.37	-0.19	-1.41	-1.52
SFO	-0.92	-0.85	0.43	-0.09
SEA	-0.88	-0.53	-0.49	-0.69
attr(,"scaled:center")				
	passengers	departures	delay_rate	cancellation_rate
	34085052.500	211370.000	23.600	1.361
attr(,"scaled:scale")				
	passengers	departures	delay_rate	cancellation_rate
	9.802957e+06	8.989402e+04	3.272783e+00	3.619530e-01

Experiment 2 — Standardization Checks and Operational Distance Mat

Scaling center and scale used by standardization
Checks after standardization Most similar pair

passengers	departures	delay_rate	cancellation_rate
0	0	0	0
passengers	departures	delay_rate	cancellation_rate
1	1	1	1

Operational distance matrix D_ops

```

      BOS  JFK  ATL  MIA  ORD  DFW  DEN  LAX  SFO  SEA
BOS  0.00  1.57  4.10  2.35  2.38  3.76  2.87  2.68  1.06  1.06
JFK  1.57  0.00  4.03  2.24  2.38  3.21  3.07  3.21  1.85  2.21
ATL  4.10  4.03  0.00  4.85  1.95  3.58  1.65  2.37  3.93  3.43
MIA  2.35  2.24  4.85  0.00  3.07  2.81  3.42  4.21  1.64  2.77
ORD  2.38  2.38  1.95  3.07  0.00  2.37  0.76  2.15  2.21  2.00
DFW  3.76  3.21  3.58  2.81  2.37  0.00  2.54  4.44  3.27  3.83
DEN  2.87  3.07  1.65  3.42  0.76  2.54  0.00  2.19  2.54  2.28
LAX  2.68  3.21  2.37  4.21  2.15  4.44  2.19  0.00  2.74  1.80
SFO  1.06  1.85  3.93  1.64  2.21  3.27  2.54  2.74  0.00  1.15
SEA  1.06  2.21  3.43  2.77  2.00  3.83  2.28  1.80  1.15  0.00

```

Experiment 2 — Differences, Eigenvalues, and MDS Coordinates

Most different and same pair

```

      row col
DEN    7    5
ORD    5    7
[1] 0.7647267

      row col
MIA    4    3
ATL    3    4
[1] 4.845267

```

Eigenvalues of B_ops

```
[1] 19.435713 13.058049 2.660368 0.845870 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
```

Operational MDS coordinates X_ops_mds

	MDS1	MDS2
BOS	-1.13	-1.03
JFK	-1.19	-0.16
ATL	2.66	0.34
MIA	-2.12	0.76
ORD	0.77	0.37
DFW	-0.12	2.57
DEN	1.22	0.46
LAX	1.33	-1.60
SFO	-1.12	-0.49
SEA	-0.30	-1.23

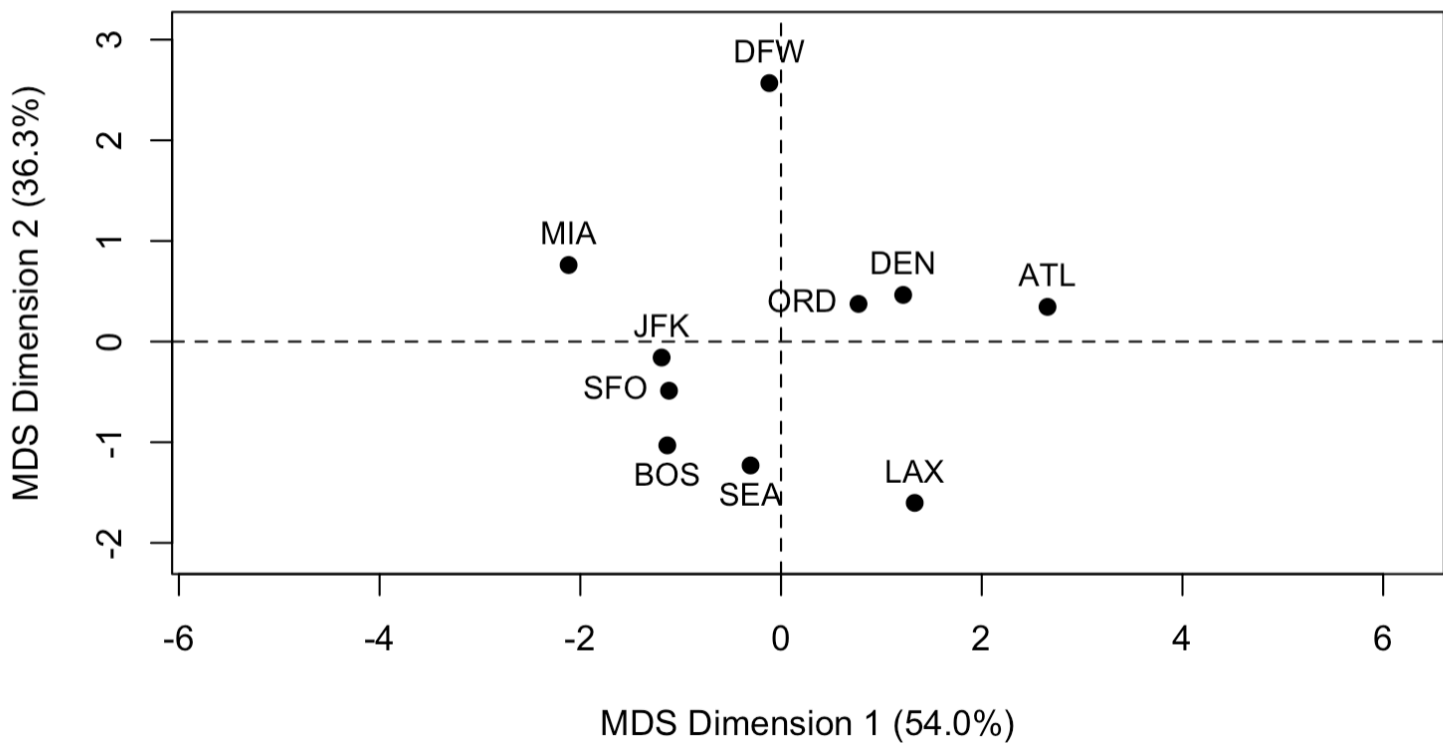
Proportion retained

```
[1] 0.5399 0.3627 0.0739 0.0235  
[1] 0.9026
```

Experiment 2 — Operational Similarity Plot

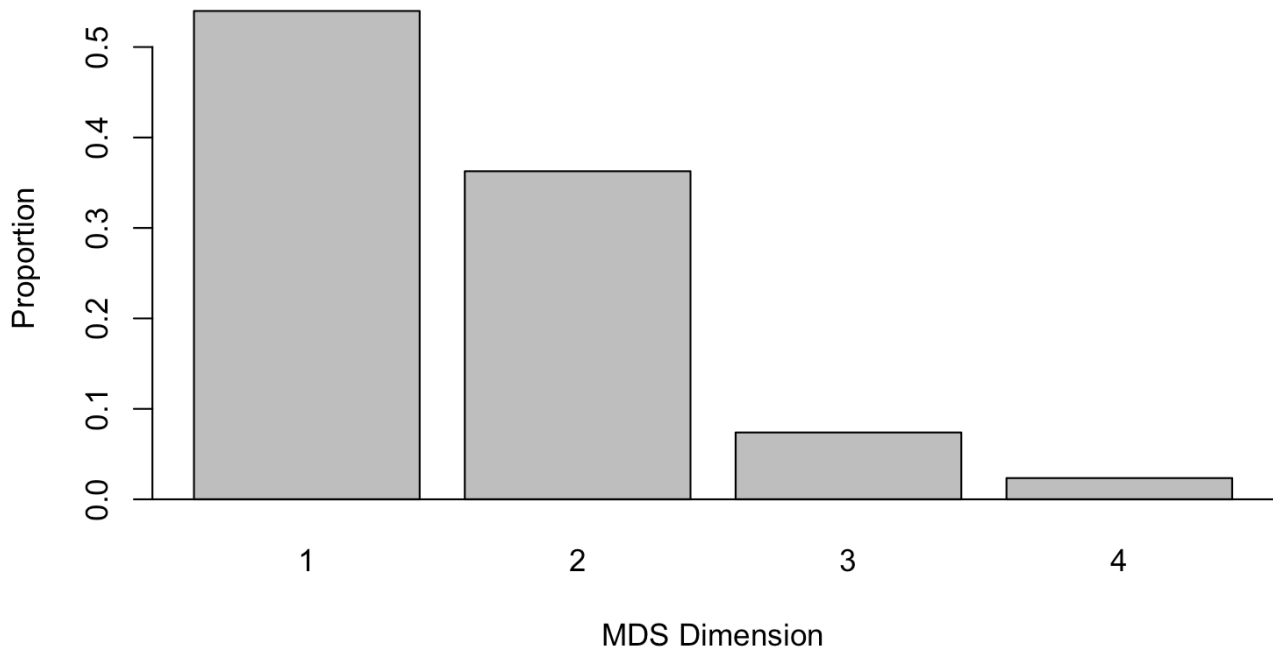
Operational similarity of 10 U.S. airports

Operational Similarity of 10 U.S. Airports



Information retained by each MDS dimension

Information Retained by Each MDS Dimension



Correlations with MDS dimensions

	MDS1	MDS2
passengers	0.85	0.46
departures	0.82	0.52
delay_rate	-0.52	0.78
cancellation_rate	-0.70	0.60

Official References

1. Federal Aviation Administration. (2025). CY 2024 passenger boarding data: Enplanements at all commercial service airports (by rank). U.S. Department of Transportation.

https://www.faa.gov/airports/planning_capacity/passenger_allcargo_stats/passenger/arp-cy2024-commercial-service-enplanements.pdf

2. Bureau of Transportation Statistics. (2024). Understanding the reporting of causes of flight delays and cancellations. U.S. Department of Transportation.

<https://www.transtats.bts.gov/HomeDrillChart.asp>

3. Bureau of Transportation Statistics. (n.d.). Airline on-time statistics and delay causes. U.S. Department of Transportation.

https://www.transtats.bts.gov/OT_Delay/OT_DelayCause1.asp

Presentation Video

<https://drive.google.com/file/d/1Fi0rqliVERrApf6nBbLT3LrDrmRYQmEI/view?usp=sharing>